

Task 9 Output

LS1_resnet_5_kmeans, k=10, label = 0

```
K = 10 most similar labels by LS1_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  Faces          Distance:  0
2.      Label:  Faces_easy      Distance:  0.000657288055061378
3.      Label:  yin_yang        Distance:  0.017300378174211373
4.      Label:  pizza           Distance:  0.018479955385930613
5.      Label:  panda           Distance:  0.01849288337451649
6.      Label:  garfield        Distance:  0.018656414298127033
7.      Label:  snoopy          Distance:  0.018668950251181315
8.      Label:  cellphone       Distance:  0.018695814085765483
9.      Label:  pigeon          Distance:  0.019778714057928415
10.     Label:  stapler         Distance:  0.01985385507274917
```

LS1_resnet_5_kmeans, k=10, label = 20

```
K = 10 most similar labels by LS1_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  beaver          Distance:  0
2.      Label:  wild_cat        Distance:  1.830917280631894e-05
3.      Label:  brontosaurus     Distance:  0.00010988608283069023
4.      Label:  dolphin         Distance:  0.00012809186763518277
5.      Label:  pigeon          Distance:  0.00014607009693834438
6.      Label:  crocodile        Distance:  0.00017346163800346925
7.      Label:  dalmatian       Distance:  0.00021224326636637425
8.      Label:  cup              Distance:  0.00022896406784289614
9.      Label:  rooster          Distance:  0.00026547012484723354
10.     Label:  platypus        Distance:  0.00027261899958230806
```

LS1_resnet_5_kmeans, k=10, label = 100

```
K = 10 most similar labels by LS1_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  yin_yang        Distance:  0
2.      Label:  snoopy          Distance:  3.3391275384264674e-05
3.      Label:  garfield        Distance:  0.00011951192018688772
4.      Label:  dollar_bill     Distance:  0.00025879806960782403
5.      Label:  scissors        Distance:  0.00026000176945217834
6.      Label:  cellphone       Distance:  0.000342910249433781
7.      Label:  stapler         Distance:  0.00046018947080017636
8.      Label:  wrench          Distance:  0.0006394054359869505
9.      Label:  headphone       Distance:  0.0007977145575985833
10.     Label:  minaret         Distance:  0.0008151468970692965
```

LS3_resnet_5_kmeans, k=10, label = 0

```
K = 10 most similar labels by LS3_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  Faces      Distance:  0
2.      Label:  Faces_easy Distance:  0.00013575609883664796
3.      Label:  soccer_ball Distance:  0.0002096563868142498
4.      Label:  euphonium  Distance:  0.0009221555252149072
5.      Label:  saxophone   Distance:  0.0011558723112041758
6.      Label:  watch      Distance:  0.0016089721936273227
7.      Label:  laptop     Distance:  0.0019271661836816056
8.      Label:  chair      Distance:  0.0024772260109335864
9.      Label:  accordion  Distance:  0.0025283979306390814
10.     Label:  umbrella   Distance:  0.002887273941816426
```

LS3_resnet_5_kmeans, k=10, label = 20

```
K = 10 most similar labels by LS3_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  ceiling_fan Distance:  0
2.      Label:  yin_yang    Distance:  0.00023649631131850146
3.      Label:  gramophone  Distance:  0.0003880147195332029
4.      Label:  dollar_bill Distance:  0.00041926222514443445
5.      Label:  scissors    Distance:  0.0012753603494212173
6.      Label:  metronome   Distance:  0.0012781900588283435
7.      Label:  mandolin    Distance:  0.0014414405358855742
8.      Label:  snoopy      Distance:  0.001796495773341844
9.      Label:  wrench      Distance:  0.003199129285612834
10.     Label:  trilobite   Distance:  0.003298032280808938
```

LS3_resnet_5_kmeans, k=10, label = 100

```
K = 10 most similar labels by LS3_resnet_5_kmeans latent semantics using distance: cosine
1.      Label:  yin_yang    Distance:  0
2.      Label:  dollar_bill Distance:  0.0002329849308065901
3.      Label:  ceiling_fan Distance:  0.00023649631131850146
4.      Label:  gramophone  Distance:  0.0005797652794052865
5.      Label:  scissors    Distance:  0.0011189977274356222
6.      Label:  snoopy      Distance:  0.0016568420197341505
7.      Label:  mandolin    Distance:  0.0017187561595993772
8.      Label:  metronome   Distance:  0.0018163860891786676
9.      Label:  wrench      Distance:  0.003495097090496113
10.     Label:  menorah     Distance:  0.003562250430394598
```